

House Martell

Vanguard's New Digital Interface

A/B Test Analysis on Client Completion Rates

Zina

Véronique

Carlos

Presentation Structure

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Introduction

Experiment:

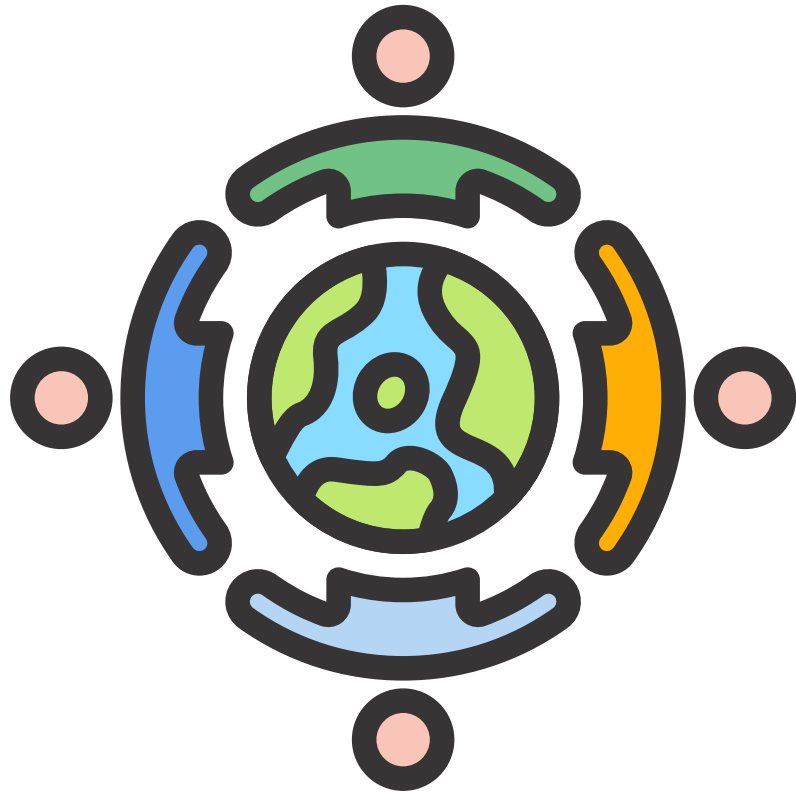
The Customer Experience team tested a new digital interface to improve online process completion.

Business Question:

Did the new UI design lead to higher process completion rates and a better user experience?

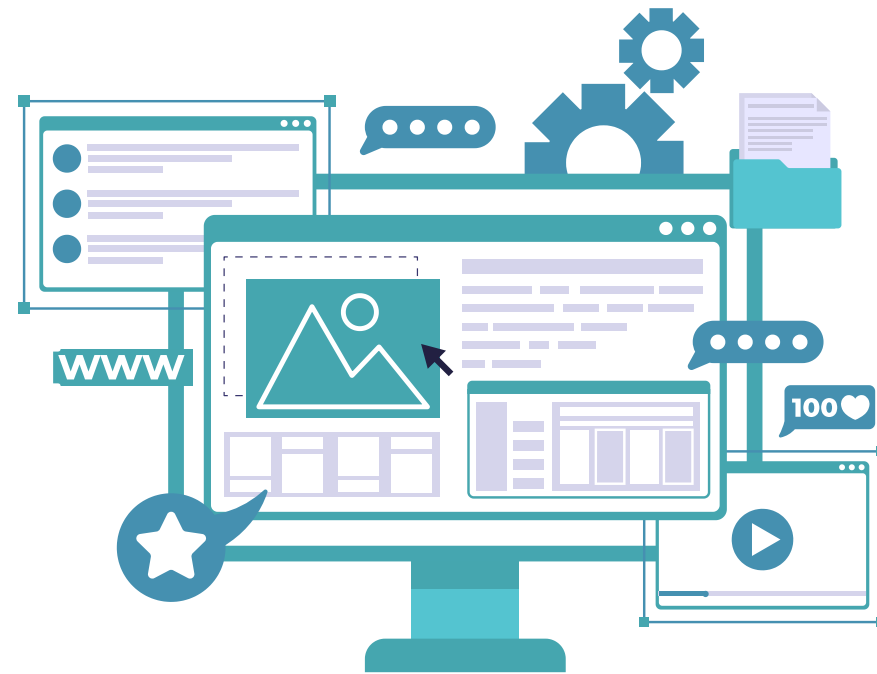
Data Overview

Three Core Datasets:



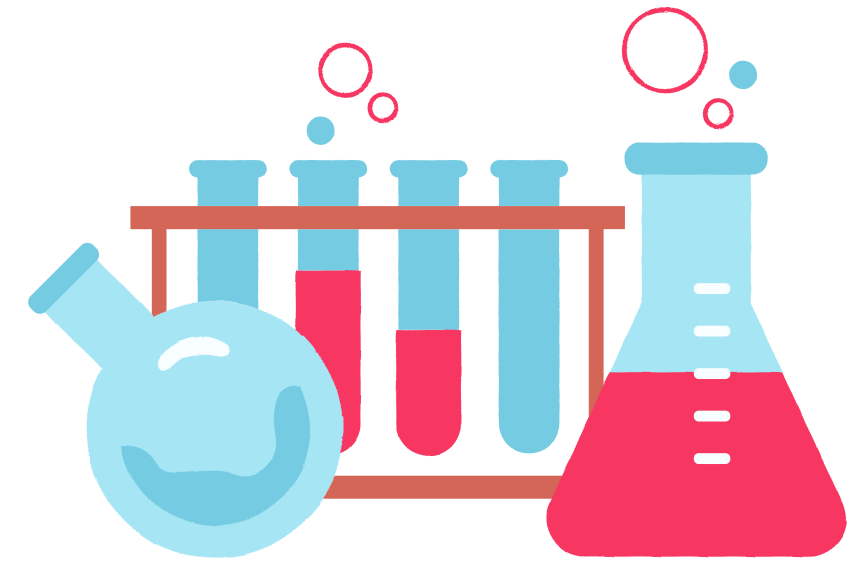
Client Profiles – 70,609 clients

- Demographics
- Financial indicators



Digital Footprints – 755,405 interactions

- User journey:
- Timestamps and visits for behavioral analysis



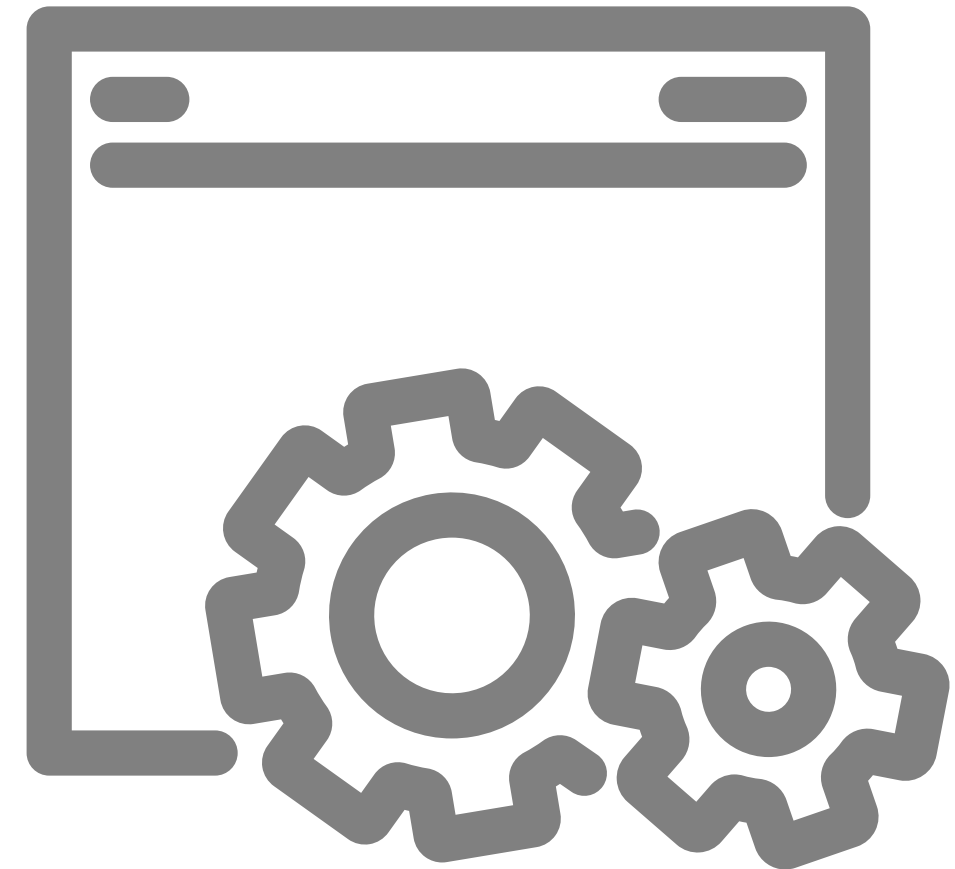
Experiment Roster – 50,500 participants

- Test group: 26,968 clients
- Control group: 23,532 clients

Data Overview

Data Preparation

- Cleaned missing and inconsistent values
- Standardized gender and age groups
- Merged datasets using client_id
- Created analysis-ready tables for KPIs



Exploratory Data Analysis (EDA)

Client Demographics

- **Age Distribution:**
 - Average Age: 46.44
- **Gender Split:**
 - Male: 33.6%
 - Female: 32.21%
- **Client Engagement:**
 - Average Tenure: 12 years
 - Digital Activity: 5.7 logins per 6 months
 - Account Holders: Majority have 2 accounts



Exploratory Data Analysis (EDA)

Client Behavior Insights

- Users spend increasing time as they move through steps
- The confirmation step takes the longest time
- Some users move backward between steps → potential confusion





Performance Metrics (KPIs)

Test vs Control

01

Completion Rate

02

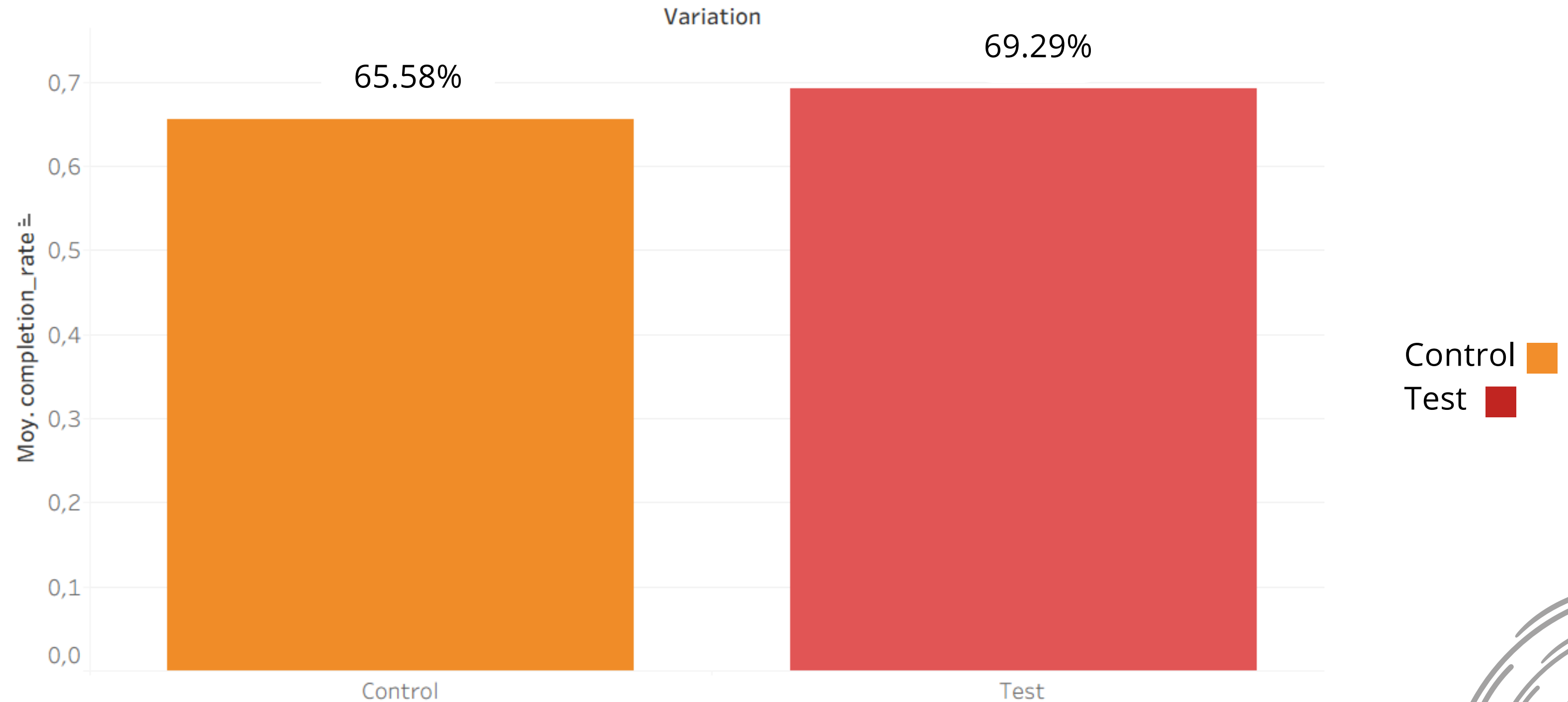
Time Spent on Each Step

03

Error Rate

Completion Rate

Completion Rate

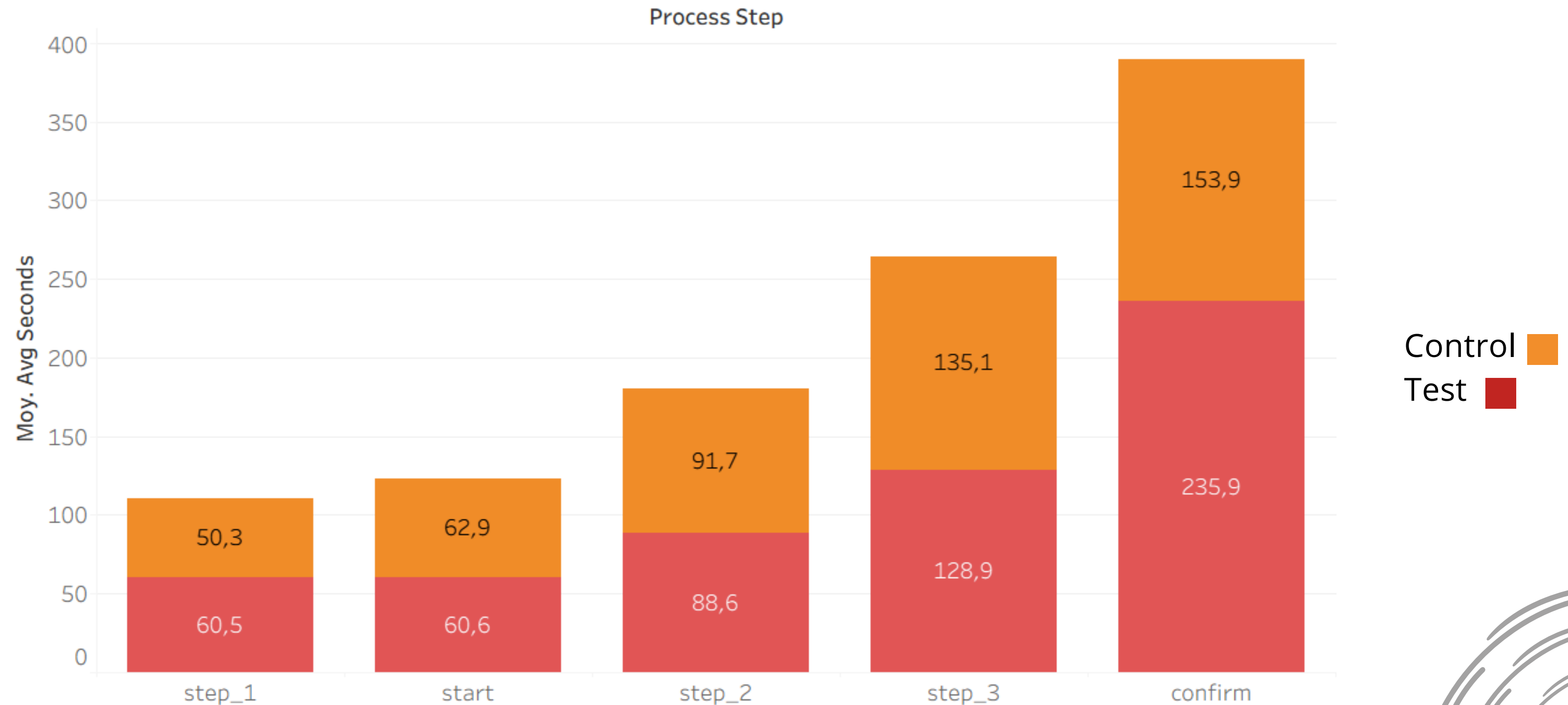


KPI Comparison: Test vs Control

- Test group completed more often than Control

Time Spent

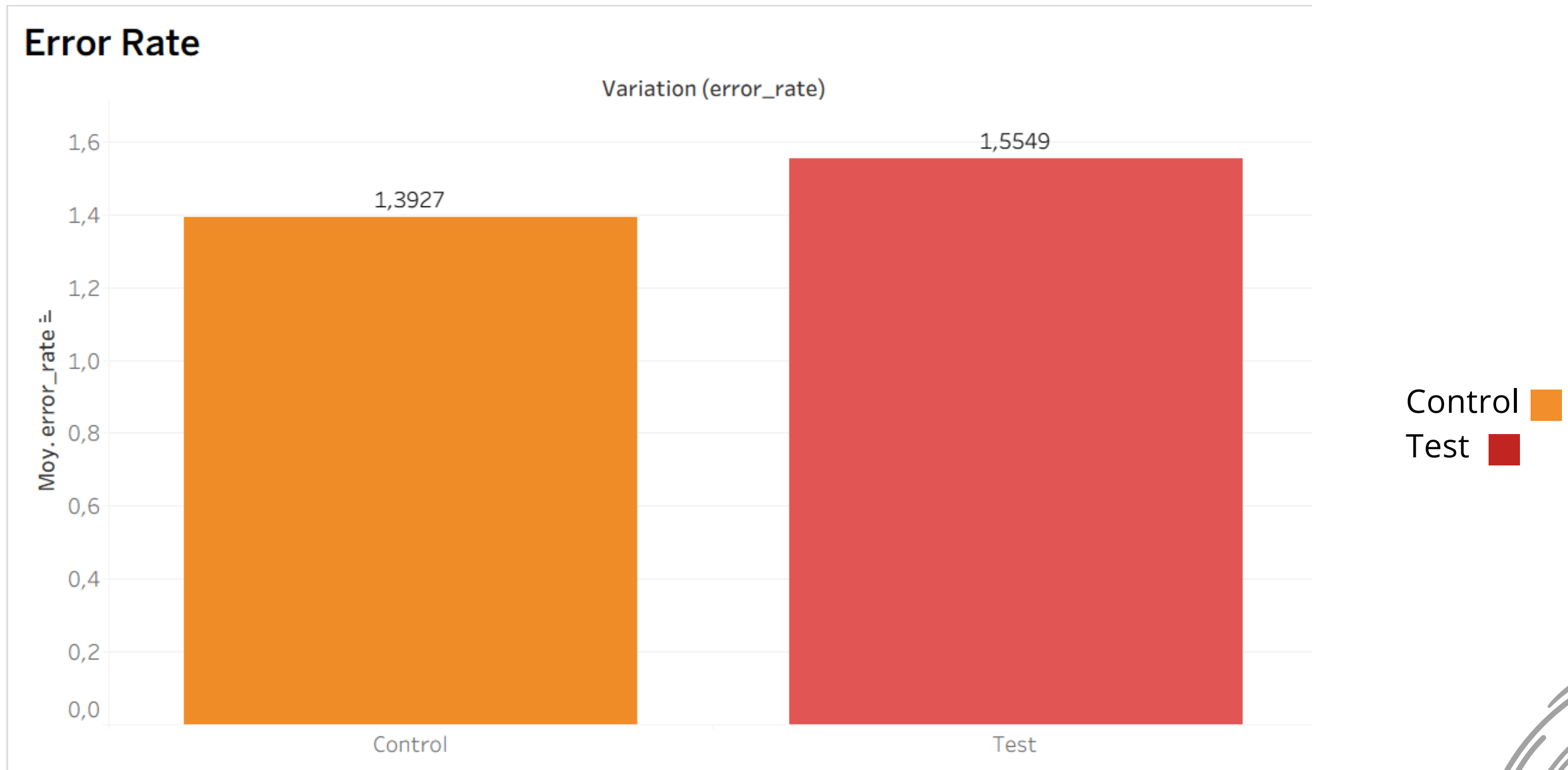
Time per Step



KPI Comparison: Test vs Control

- Test users generally moved faster through steps
- Confirmation step still a friction point

Error Rate



KPI Comparison: Test vs Control

- Error Rate
- Test group showed slightly higher navigation errors

Hypothesis Testing

The Goal

to see if the new design leads to a better user experience and higher process completion rates

Hypothesis

- Null H_0 – No difference between Test and Control
- Alternative H_1 – Test Completion Rate is Higher

p-value $\ll$ 0.05

Hypothesis Testing

Cost-Effectiveness Threshold

Vanguard requires an absolute increase of at least 5 % points in completion rate to justify costs.

Observed Result

- Control: 65.6%
- Test: 69.3%
- Absolute increase: +3.7 % points
- Does NOT meet Vanguard's cost-effectiveness threshold



Hypothesis Testing

Additional Hypothesis → Client Age

Tested whether average age differs between Test and Control

Result: $p = 0.016$

**Test used:
two-sample t-test**

Client Age T-test
Statistically significant –
Unlikely to bias main
experiment conclusions

**Average age :
Control group = 47.50
Test group = 47.16**

Experiment Evaluation

Design Effectiveness

- Clear A/B structure
- Large sample size

Duration Assessment

- 3 months duration
- Enough data for reliable statistical analysis

Additional Data That Would Help

- Device type (mobile vs desktop)
- Session abandonment reasons
- Client feedback or satisfaction scores

Challenges & Learnings



Challenges

- Managing different data
- Connecting SQL, Python, and Tableau
- Understanding hypothesis testing logic

Learnings

- Importance of data modeling
- Choosing correct statistical tests
- Translating technical results into business insights



Conclusions

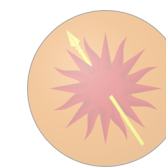
The new design improves user outcomes, but further optimization is needed before full rollout.

Key Findings

- New UI increases completion rate
- Improvement is statistically significant
- Does not yet meet cost-effectiveness threshold

Recommendations

- Improve UX, especially confirmation step
- Reduce navigation errors
- Run a follow-up experiment focused on efficiency



House Martell

Thank
You!

Zina

Véronique

Carlos

Hypothesis Testing



5.6%

Reject H_0 – Test is statistically superior

Statistical Z-test

Z-statistic: 8.87

Client Age T-test

Statistically significant –
Unlikely to bias main
experiment conclusions

ROI

Control: 65.6%
Test: 69.3%
Absolute increase: +3.7 % points